

D: Bingwhenever

Source file name: `bingwhenever.c`, `bingwhenever.cpp`, `bingwhenever.java`, or `bingwhenever.py`

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A group of N friends wants to play bingo together. Unfortunately, they cannot agree on a date, a time, or even a place to meet next Saturday. They eventually come up with a brilliant solution: they will play Bingwhenever.

Random Enterprises Developing Intelligent Systems (REDIS) has developed Bingwhenever, a rather simple generalization of common bingo. Each friend receives a Bingwhenever card, i.e., a card with K different integer numbers on it, each one between 1 and M . Then REDIS produces a random sequence of B different numbers, each one also between 1 and M , just as the numbered balls drawn in a common bingo.

The sequence is sent to everyone by email, and each player uses a small program to follow the game and mark the numbers on his/her card as the balls are drawn. The game is won at step s of the game by those players that have marked all the numbers of his/her card when the s -th ball of the sequence is drawn, provided that nobody has done it before (at an earlier step).

In order to avoid discussions, at the end of the day another email from REDIS will announce the winner.

Your task is to help REDIS determining the winner or winners of the game.

For example, let's say Alice, Bob, and Charles are friends who are going to play Bingwhenever, and each card has 4 numbers. Alice's card is 3, 7, 10, 15; Bob's card is 3, 4, 7, 10, and Charles's card is 2, 7, 11, 15. Let's also say that the list of 8 balls is drawn in the following order: 4, 5, 7, 3, 2, 10, 11, 15. In that case, Bob is the first to complete his card and win the game (Alice marked 3 numbers and Charles only 2). But if the list of balls is 5, 7, 3, 2, 10, 11, 15, 4, then Alice and Charles complete their cards when the ball 15 is drawn, and Bob is left waiting for the ball 4.

Given the players' names, their Bingwhenever cards, and the order in which the balls are drawn, determine the names of all winners.

Input

The input consists of several test cases. Each test case starts with a line containing three integers N ($1 \leq N \leq 10^3$), K ($1 \leq K \leq 10^3$), and B ($K \leq B \leq 10^6$), representing the number of players, the number of numbers on each Bingwhenever card, and the number of balls that will be drawn (i.e., the length of the random sequence).

The next N lines describe the players. Each line contains a player's name followed by the K different integer numbers on their Bingwhenever card, each one of these between 1 and M , with $M = 10^9$. Player names consist only of lowercase English letters and contain no spaces. All player names are distinct.

The next line contains B different integers, representing the sequence of the numbered balls in the order in which they are drawn. All ball numbers are integers between 1 and M . Every number appearing in a Bingwhenever card also appears in the sequence of B drawn balls.

The last line of the input will be 0 0 0 and should not be processed.

The input must be read from standard input.

Output

For each case print a line with the names of all winners in lexicographical ascending order, separated by a single space.

The output must be written to standard output.

Sample Input	Sample Output
<pre>3 4 8 alice 3 7 10 15 charles 2 7 11 15 bob 3 4 7 10 5 7 3 2 10 11 15 4 3 4 8 alice 3 7 10 15 charles 2 7 11 15 bob 3 4 7 10 4 5 7 3 2 10 11 15 0 0 0</pre>	<pre>alice charles bob</pre>